Raisa Nusrat

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Research Interests

Deep learning for medical imaging; facial analysis for neurological disorder detection (stroke, ALS, Parkinson's disease); multimodal fusion of imaging and clinical data; explainable and generalizable healthcare AI; bio-inspired robotics and autonomous drones.

Education

B.Sc. in Electrical and Electronic Engineering 2020 – June 2026
Bangladesh University of Engineering and Technology (BUET), Dhaka CGPA: 3.28/4.00
Thesis: *A Multimodal Deep Learning Framework for Stroke Detection from Facial Asymmetry and Clinical Data with Explainable AI and External Validation* — Supervisor: Dr. Mohammad Faisal, Professor, EEE, BUET

Publications

Journal Articles

1. **R. Nusrat**, M. R. Rahman, I. J. Riya, and M. Faisal, "Age-Robust Stroke Detection and Paretic-Side Identification Using Nasolabial Fold Asymmetry and Clinical Data," *IEEE Journal of Biomedical and Health Informatics*, under review, 2026.
2. **R. Nusrat**, M. S. Raihan, S. M. R. Rahman, H. Saha, and S. S. Ahbab, "A Low-Cost Integrated Motorcycle Theft Detection and Monitoring System Using Arduino, ESP32-CAM, and GSM/GPS Technologies," *Journal of the Electrical and Electronic Engineering Division, Institution of Engineers, Bangladesh (IEB)*, accepted, 2026.

Conference Papers

3. **R. Nusrat**, "When Scale Flips the Answer: Corridor-Level Deep Reinforcement Learning for Traffic Signal Control on a Simulated Dhaka Arterial," *AABEA 2026 Biennial Convention*; abstract accepted for proceedings, full paper under review.
4. **R. Nusrat**, "Rare-Attack Blindness in Federated Intrusion Detection for the Internet of Medical Things," *14th International Conference on Electrical and Computer Engineering (ICECE 2026)*, Dhaka, under review.

Manuscripts in Preparation

5. **R. Nusrat** et al., "Differentiating stroke from ALS using facial-motion asymmetry on the Toronto NeuroFace dataset."
6. **R. Nusrat** et al., "Parkinson's disease detection using the PADS smartwatch dataset."
7. **R. Nusrat**, A. Hamim, M. Fahim, M. J. Hasan, and A. Hasan, "HIVE-R: Bio-inspired and visual odometry based drone for rescue operations in GPS-denied environments."

Research Experience

Undergraduate Thesis Research — Stroke Detection 2025 – 2026
Department of EEE, BUET | Supervisor: Dr. Mohammad Faisal

- Built a clinical dataset of 117 patients (71 stroke, 46 non-stroke) from hospital records, with facial images and 38 laboratory features.
- Designed a Siamese CNN (EfficientNet-B0) comparing left and right facial regions; combined image AUC of 0.983.
- Fused image and clinical models; 80% accuracy and 88.9% sensitivity on 15 held-out external patients.

Clinical Study — Multimodal Stroke Detection (Facial Imaging and Blood Biomarkers) 2026 – Present

- Wrote the study protocol and obtained BUET Ethical Review Committee approval (Ref. BUET/ERC/2026/01).
- Supported by the Basic Research Grant, RISE, BUET.

Independent Research — Deep Reinforcement Learning for Traffic Signal Control 2026

- Built a five-junction Dhaka corridor in SUMO and trained a shared-policy DQN; under saturation, cut arterial delay by 60% and cross-street delay by 41% versus fixed-time control.

Robotics Research — HIVE-R Autonomous Rescue Drone 2025 – Present

- Honeybee-inspired drone swarm for search and rescue in GPS-denied environments.
- Camera-IMU visual odometry (SLAM) on Pixhawk and Raspberry Pi; YOLOv8-Nano human detection with fuzzy augmentation for smoke, blur, and low light.

Selected Projects

Multi-Modal Parkinson's Disease Detection EEE 402, AI & ML Lab
Detection from speech, gait, and handwriting with late fusion.

Motorcycle Theft Detection and Monitoring System Embedded Systems
Arduino and ESP32-CAM unit with PIN-locked ignition, live camera view, and GPS/SMS tracking; working prototype at about 3800 BDT.

Autonomous Trash-Collecting Line-Following Robot EEE 318, Control Systems Lab
PID-controlled line follower with object detection and collection.

Noise Cancellation Using Adaptive Filters EEE 312, DSP Lab
Compared LMS and RLS adaptive filters by SNR and mean squared error.

Microgrid Simulation with Intentional Islanding EEE 306, Power Systems Lab
Simulink model of a neighbourhood microgrid that islands from the grid during faults.

Teaching Experience

Mathematics Tutor (International) 2026 – Present
Teaching students in Australia on the Edexcel curriculum; personalized lesson plans and progress tracking.

Undergraduate Academic Consultant (International) 2024 – 2025
Online instruction for EEE and Mathematics students at Stony Brook University, USA: circuit analysis, signals and systems, and advanced calculus.

Content Developer and Academic Support, Udvash Academic & Admission Care 2022 – 2023
Wrote conceptual questions and parallel texts; provided real-time online Q&A support.

Private Tutor (National Curriculum) 2022 – Present
Physics and Mathematics for Classes 9–12 (SSC/HSC).

Industrial Training

Industrial Attachment, Bangladesh Television (BTV)
Satellite communication, high-power RF transmission, and studio broadcasting electronics; hands-on signal acquisition and hardware operation.

Awards and Activities

- Champion, Integration Bee, organized by Tachyon.
- Participant, Robotics Workshop.

Technical Skills

Programming: Python, MATLAB, C/C++, LaTeX
Machine Learning: PyTorch, scikit-learn, OpenCV, YOLOv8
Design and Simulation: MATLAB/Simulink, SUMO, PSpice, Proteus, Verilog, Quartus, AutoCAD
Hardware: Arduino, ESP32-CAM, Raspberry Pi, Pixhawk (MAVLink), IMU, GPS/GSM modules

Test Scores

TOEFL iBT: scheduled 14 November 2026